

CtrlHGen: 可控逻辑假设生成 与复现进展

从论文机制到可审计实验链路

论文: 可控溯因 = 执行语义 + 条件遵循
复现: 监督契约 → 实体覆盖 → GRPO 组内信号

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Observation

Systemic Lupus Erythematosus, Antiphospholipid Syndrome, Sjögren's Syndrome

Hypothesis 1:

$H = V_? : \text{Produce}(V_?, \text{Autoantibodies}) \wedge \text{Property}(V_?, \text{Chronic})$
 $\wedge \text{Property}(V_?, \text{Autoimmune})$

Interpretation:

The disease that produces autoantibodies and is chronic and autoimmune.

Hypothesis 2:

$H = V_? : \text{Treat}(\text{Hydroxychloroquine}, V_?) \wedge \text{Property}(V_?, \text{Autoimmune})$

Interpretation:

The disease that are autoimmune and can be treated with hydroxychloroquine.

Hypothesis 3:

$H = V_? : \text{Affected}(V_?, P_?) \wedge \text{Gender}(P_?, \text{Female})$
 $\wedge \text{Age}(P_?, \text{Reproductive}) \wedge \text{Carries}(P_?, \text{HLA-DR3})$

Interpretation:

The disease that is more likely to occur in women of reproductive age who carry the HLA-DR3 susceptible allele.

Pathology



Treatment



Susceptibles



(a) Control Semantic Domain

Observation

Chris Paul, James Harden, Russel Westbrook

Hypothesis 1:

$H = V_? : \text{PlayedFor}(V_?, \text{Thunder}) \wedge \text{Won}(V_?, \text{AssistTitle})$

Interpretation:

Players who have played for the Oklahoma City Thunder and have won the NBA assist title.

Hypothesis 2:

$H = V_? : \text{PlayedFor}(V_?, \text{Thunder}) \wedge \text{Won}(V_?, \text{AssistTitle})$
 $\wedge \neg \text{Won}(V_?, \text{NBACHampionship})$

Interpretation:

Players who have played for the Thunder, have won the NBA assist title, and have never won an NBA championship.

Hypothesis 3:

$H = V_? : \text{PlayedFor}(V_?, \text{Thunder}) \wedge \text{Won}(V_?, \text{AssistTitle})$
 $\wedge \neg \text{Won}(V_?, \text{NBACHampionship}) \wedge \text{selected}(V_?, \text{ALL_Star})$

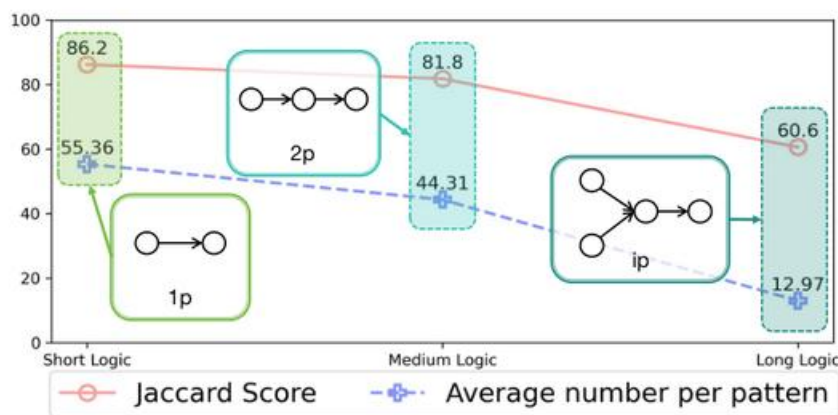
Interpretation:

Players who have played for the Oklahoma City Thunder, have won the NBA assist title, have never won an NBA championship, but have been selected as NBA All-Stars.



(b) Control Structure Complexity

同一观察 O 可以有多种合理 H；控制 C 决定语义焦点或结构复杂度。



(a) Hypothesis Space Collapse



(b) Hypothesis Reward Oversensitivity

长逻辑：候选 55.36 $\rightarrow$ 12.97, Jaccard 86.2 $\rightarrow$ 60.6 | 单谓词错误：Jaccard = 2/45 = 0.044

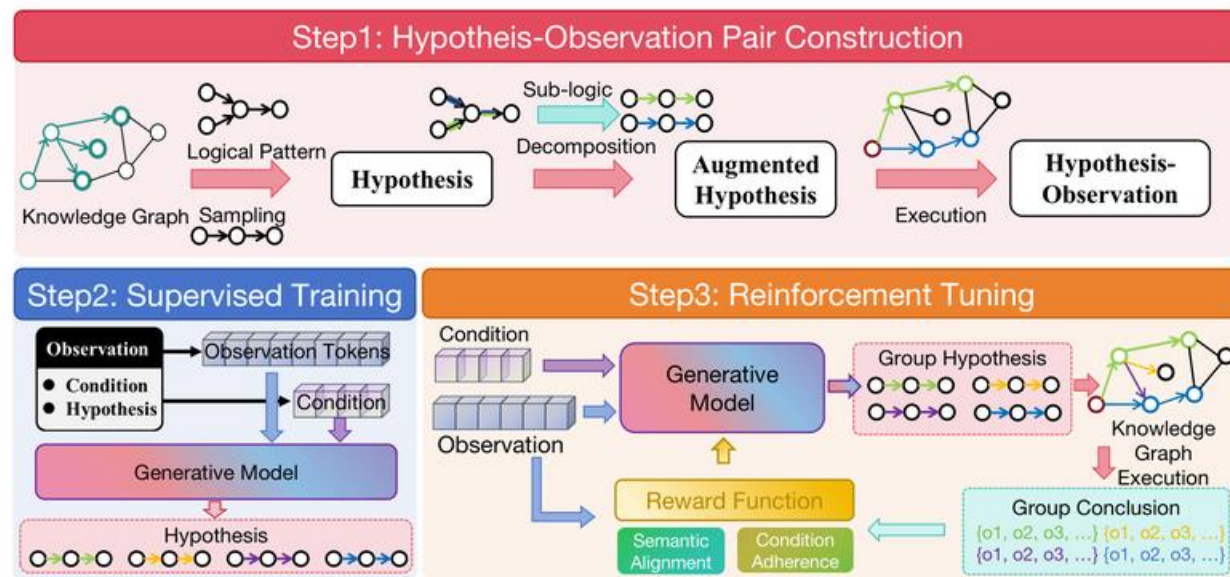


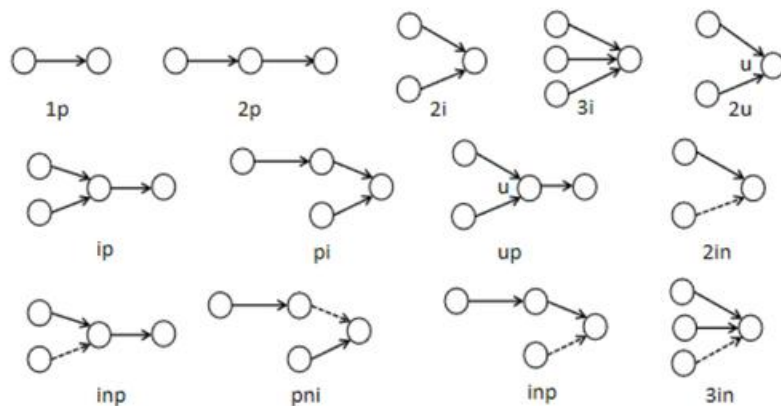
Figure 3: An overview of our controllable abductive reasoning framework. The process consists of three main steps: (1) Hypothesis-Observation pair construction through sub-logic decomposition to expand the hypothesis space, (2) Supervised training of the generative model using augmented hypotheses, and (3) Reinforcement tuning with dual rewards for semantic alignment and condition adherence to balance hypothesis accuracy with control signal compliance.

数据构造与分解 → 两阶段 SFT → KG 执行反馈与 GRPO。

课程式增广：从复杂结构抽取同源子任务

inp ————— $2p \rightarrow O_A$
 $2p \rightarrow O_B$

不是任意造样本：子逻辑来自同一个复杂假设，
并在 KG 上重新执行得到子观察。



五类控制：pattern · 实体数 · 关系数
指定实体 · 指定关系

论文只分解 up / 3in / pni / pin / inp 五类复杂 pattern。

Reward definition

$$\begin{aligned} R_{\text{sem}} &= \text{Jaccard} + 0.5 \text{ Dice} \\ &\quad + 0.5 \text{ Overlap} \\ R_{\text{cond}} &= 1[H \text{ satisfies } C] \\ R &= 0.5 R_{\text{sem}} + 0.5 R_{\text{cond}} \end{aligned}$$

- 同一 prompt 采样 $k = 4$ 个候选
- 组内标准化相对奖励；KL 约束策略漂移
- 关键是组内能否形成可排序差异

Table 3: Results of ablation studies for the reward function.

Model	Semantic Similarity			Condition Adherence		Average
	Jaccard	Dice	Overlap	Accuracy	Smatch	
CtrlHG(w/o RL)	71.5 $\pm$ 0.37	75.8 $\pm$ 0.35	83.7 $\pm$ 0.33	81.5 $\pm$ 0.38	79.0 $\pm$ 0.18	78.3
CtrlHG(w/o Dice and Overlap)	74.8 $\pm$ 0.34	78.2 $\pm$ 0.33	85.1 $\pm$ 0.30	90.3 $\pm$ 0.25	82.0 $\pm$ 0.15	82.1
CtrlHG(w/o Condition Adherence)	77.5 $\pm$ 0.33	81.6 $\pm$ 0.31	87.8 $\pm$ 0.29	68.3 $\pm$ 0.46	75.0 $\pm$ 0.22	78.0
CtrlHG	77.0 $\pm$ 0.34	80.8 $\pm$ 0.31	86.8 $\pm$ 0.28	93.5 $\pm$ 0.24	83.3 $\pm$ 0.15	84.3

去掉条件奖励：集合语义略升，但 Accuracy 93.5 $\rightarrow$ 68.3。

Table 3 表明：语义质量与条件遵循是明确的双目标，而非单一分数。

项目	论文正文	作者代码意图	本复现
模型	12-layer GPT-2	GPT2_6 / num_layers: 6	explicit n_layer = 6
优化与 LR	AdamW / 1e-5	Adam / 5e-5	Adam(SFT), AdamW(GRPO)
训练	400 + 50 epoch	WN batch 160	50 + 50 epoch, seed 42
资源与范围	4 × A6000 / 3 KG / 5 controls	公开仓库	1 × L20 / WN18RR / pattern

定位：方法与主要现象复现，不是论文绝对数值逐点复刻。

设置不完全同分布；后续所有“论文对照”只解释尺度，不解释严格优劣。

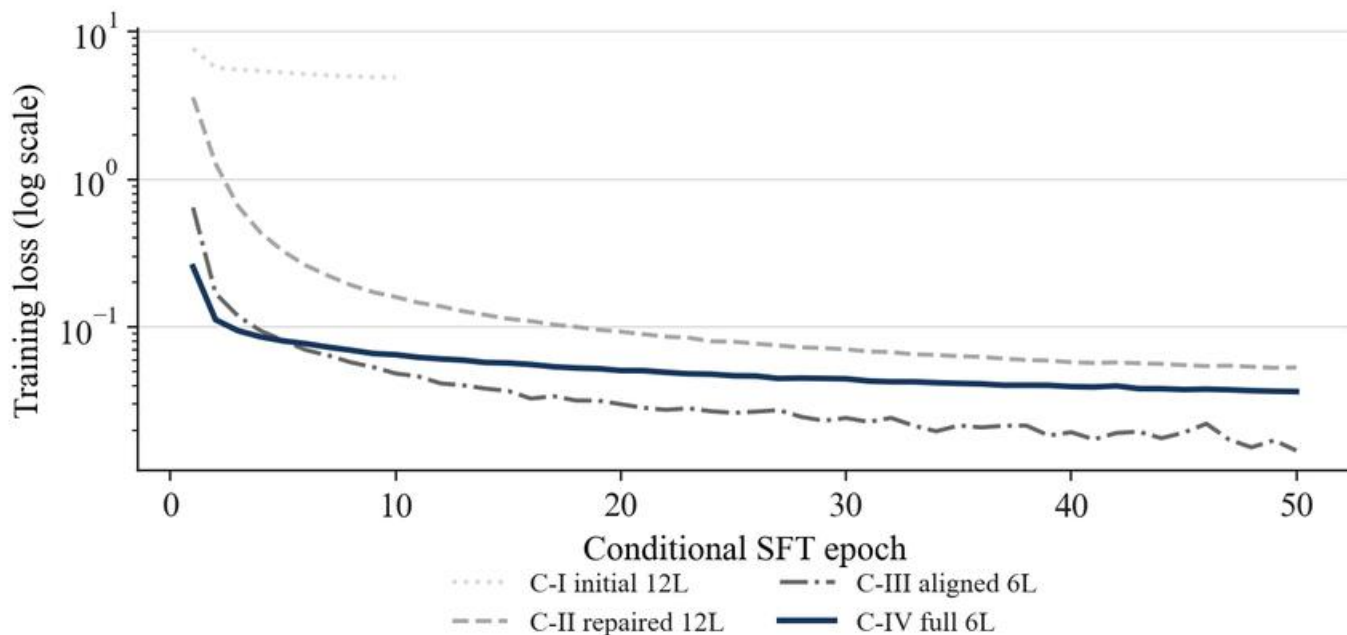
复现路线：从“能跑”到“可信”

REPRO · JOURNEY



三次关键转折：监督契约 → 实体覆盖 → GRPO 组内信号。

转折一：loss 下降，但监督实际上失效



answers | COND | pattern | SEP
target | END

active labels = target + END

根因

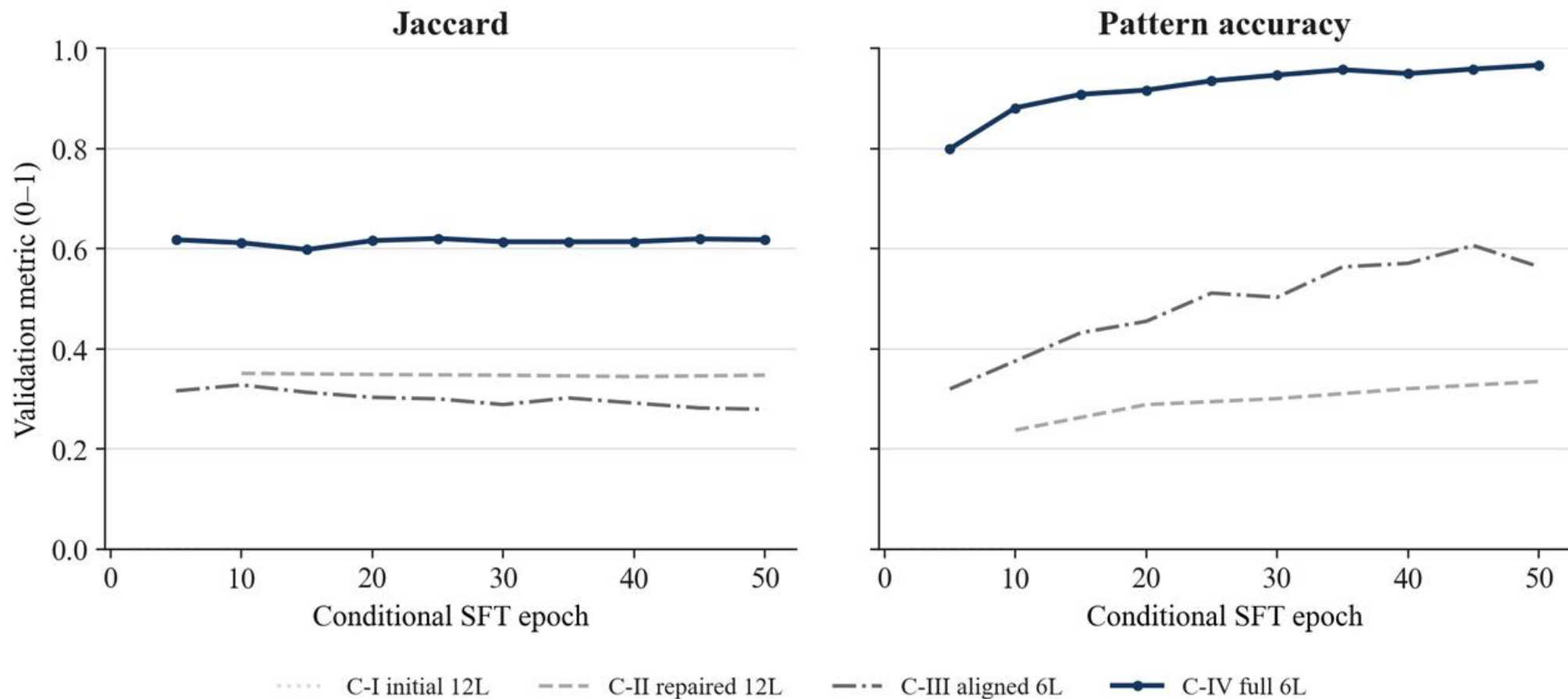
- left-padding backend 状态污染
- pair mask 错位; target / END 被屏蔽

修复

- 训练固定 right padding; 生成局部切换
- checkpoint contract + Parse / EOS gate

teacher-forced loss 不能替代自由生成健康检查。

转折二：结构控制与集合语义是两个能力维度



C-II → C-III: Pattern Accuracy 0.3185 → 0.5883, 但 Jaccard 0.3299 → 0.2690

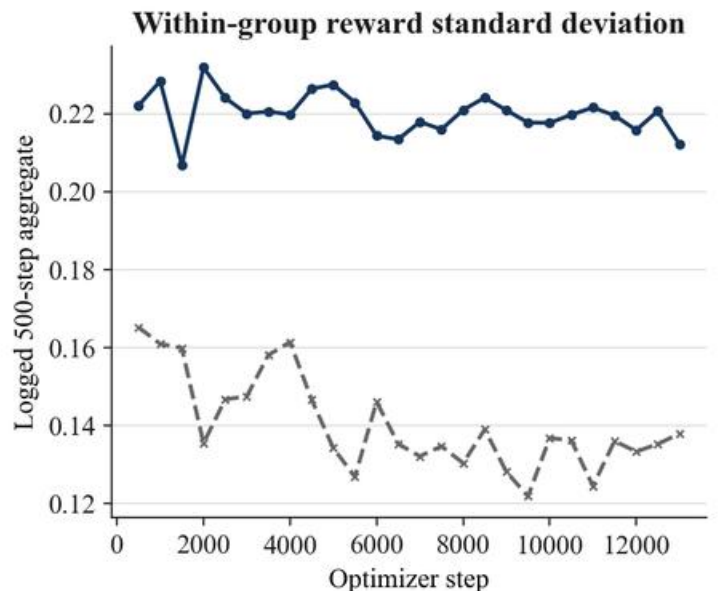
train / pattern 1,024 → 8,000 (7.81×)

指标	C-III	C-IV	变化
实体词表覆盖	35.41%	81.86%	+46.45 pp
test 未见实体 occurrence	27.70%	3.34%	-24.37 pp
greedy Jaccard	0.2690	0.5969	+0.3279
Pattern Accuracy	0.5883	0.9441	+0.3558

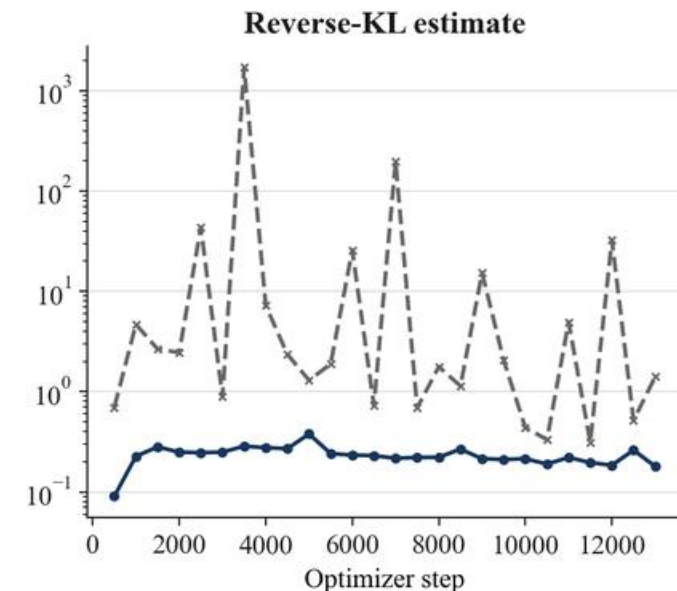
关系始终 22/22 覆盖；最直接解释是实体监督被补足。

边界：样本增多同时带来更多 optimizer steps，仍需 compute-matched 消融。

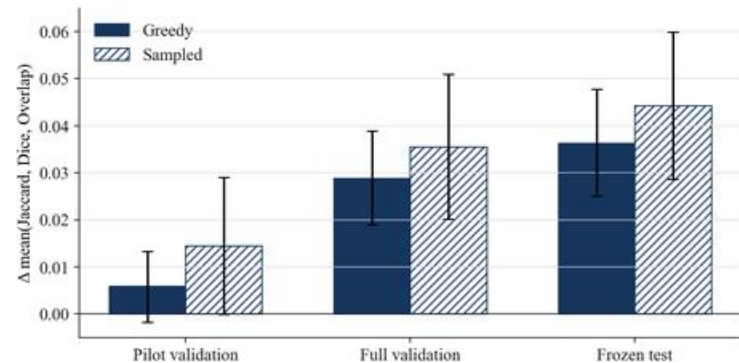
转折三：高 reward 不等于有效 GRPO 信号



--- Original Phase D



—●— Repaired full

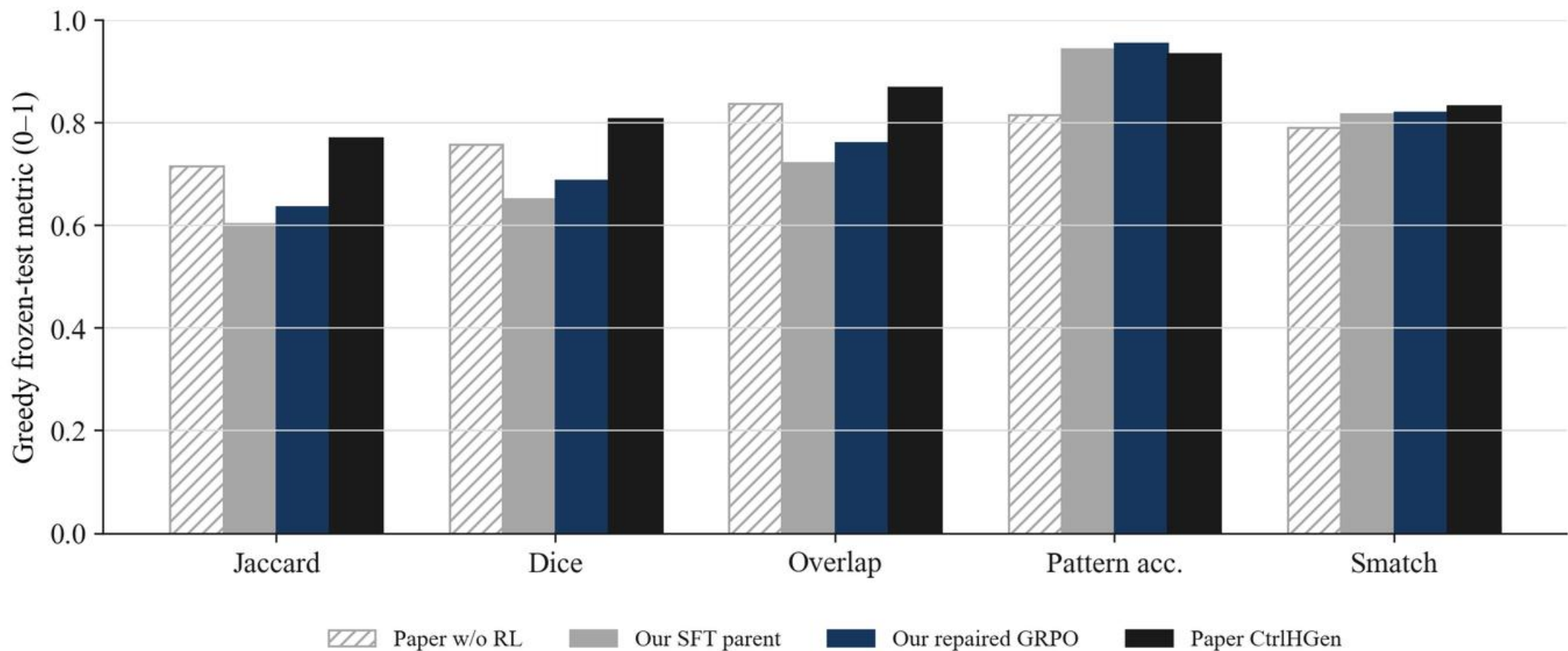


Original: 复用 SFT 数据, 缺 SEP
Repaired: fresh RL-only + 显式 SEP

repaired 版本组内 reward std 更高、KL 更稳定；full / test 配对区间均高于 0。

最终结果：主要现象已复现，绝对差距仍在

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Greedy Δ mean = +0.03625 CI [0.02506, 0.04773] | Sampled Δ mean = +0.04426 CI [0.02864, 0.05991]

绝对指标统一使用 0-1 纵轴；论文只作非同设置的尺度参照。

已经得到的证据

- pattern 控制稳定有效
- GRPO 对集合语义产生配对正增益
- 监督、覆盖与 RL 信号均可审计

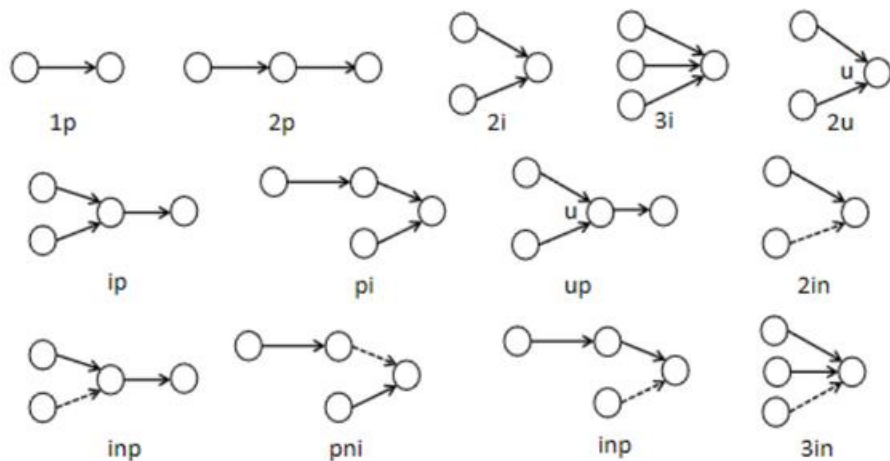
不能扩大的结论

- 不是论文绝对数值的严格复刻
- 单 seed 不是跨训练显著性
- fresh data 与 SEP 尚未拆分

下一步

- 12 层 depth-only
- 多 seed 与 fresh RL shard
- fresh × SEP 2×2
- 新 untouched test

核心方法现象已复现；结构控制已强，实体—关系实例化后的集合语义仍是主要差距。



$$\text{Jaccard} = |A \cap O| / |A \cup O|$$
$$\text{Dice} = 2|A \cap O| / (|A| + |O|)$$
$$\text{Overlap} = |A \cap O| / \min(|A|, |O|)$$

$$R_{\text{sem}} \in [0, 2]$$
$$R_{\text{cond}} \in [0, 1]$$

Overlap 在一集合包含另一集合时可为 1, 因此必须与 Jaccard / Dice 联合解读。

13 类清单来自 pattern_filtered.csv; 不包含 pattern_table.csv 中额外的 3p。

项目	论文正文	作者代码意图	公开代码实际语义	本复现
模型深度	12 层	GPT2_6 / num_layers:6	字段未覆盖 n_layer	显式 n_layer=6
Optimizer	AdamW	未写	实际使用 Adam	Adam / AdamW (GRPO)
SFT LR	1e-5	5e-5	5e-5	5e-5
Batch	256	WN:160	160	160
Epoch	400+50	50	315 resume / total 50	50+50
Warm-up	50 / 5 epoch	5 step	约 5 step	按 optimizer step
算力	4×A6000	—	—	1×L20

只陈述可验证差异；本复现采用显式、自洽、可审计的实现。

解码 / 模型	Jaccard	Dice	Overlap	PA	Smatch	Parse	EOS
Greedy · SFT parent	0.60304	0.65173	0.72159	0.94291	0.81676	0.99159	1.00000
Greedy · Repaired	0.63667	0.68702	0.76142	0.95433	0.82006	0.99219	1.00000
Sampled · SFT parent	0.55997	0.60575	0.67287	0.94351	0.81270	0.99038	1.00000
Sampled · Repaired	0.60490	0.65023	0.71625	0.95132	0.81681	0.99399	1.00000

conditional-best → epoch 50 (训练规则)
phase-d-parent → epoch 45 (研究目标)

parent bake-off 使用过同一 test split;
终点未读 test, 但端到端不是 untouched holdout.

paired bootstrap: 固定模型在 1,664 条记录上的重采样, 不是跨 seed 方差。